

Trabajo de Fin de Máster

Centelleo en Mezclas Gaseosas Ar – CF₄ & Ar – N₂

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1. Introducción

2. Objetivos

3. Espectros Experimentales

3.1. Ar

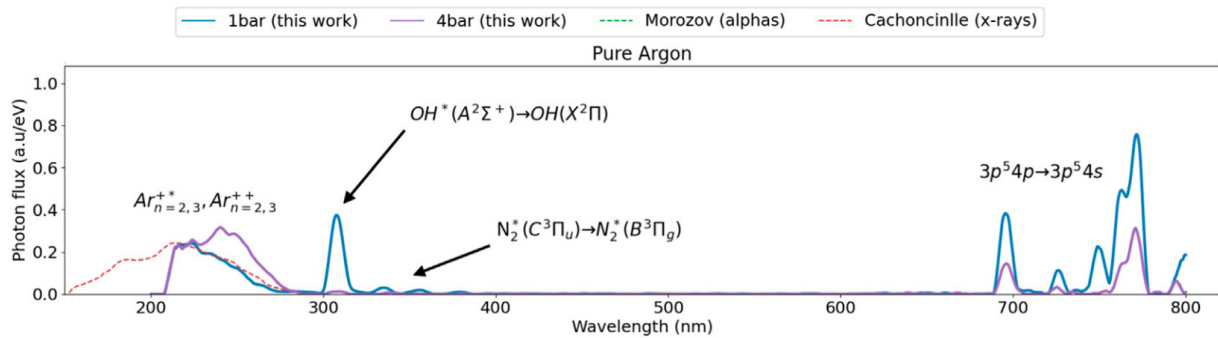


Figura 1: Caption

3.2. CF₄

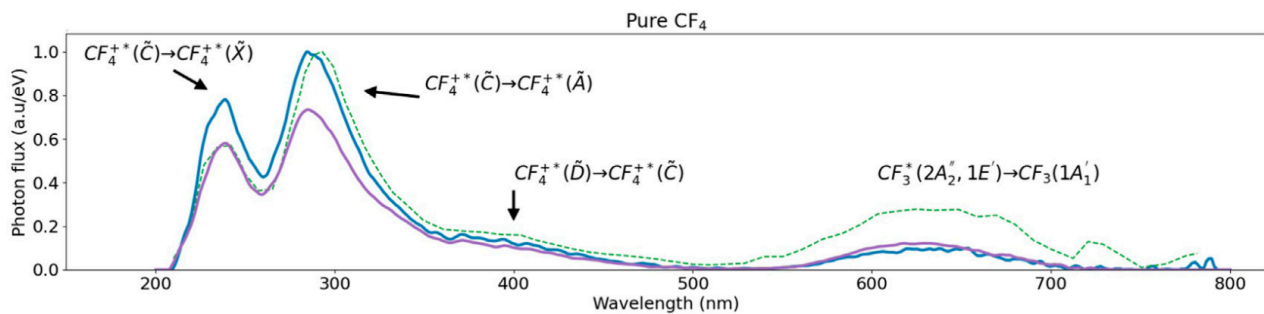


Figura 2: Caption

3.3. N₂

3.4. Ar – CF₄

3.5. Ar – N₂

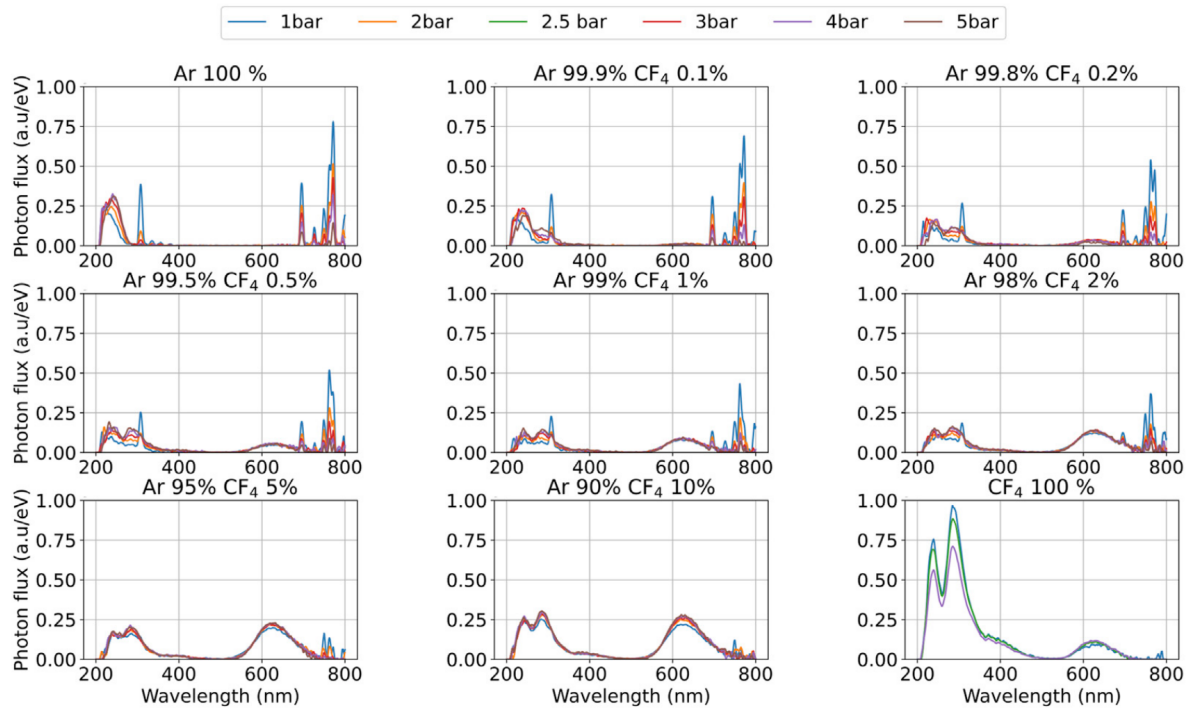


Figura 3: Caption

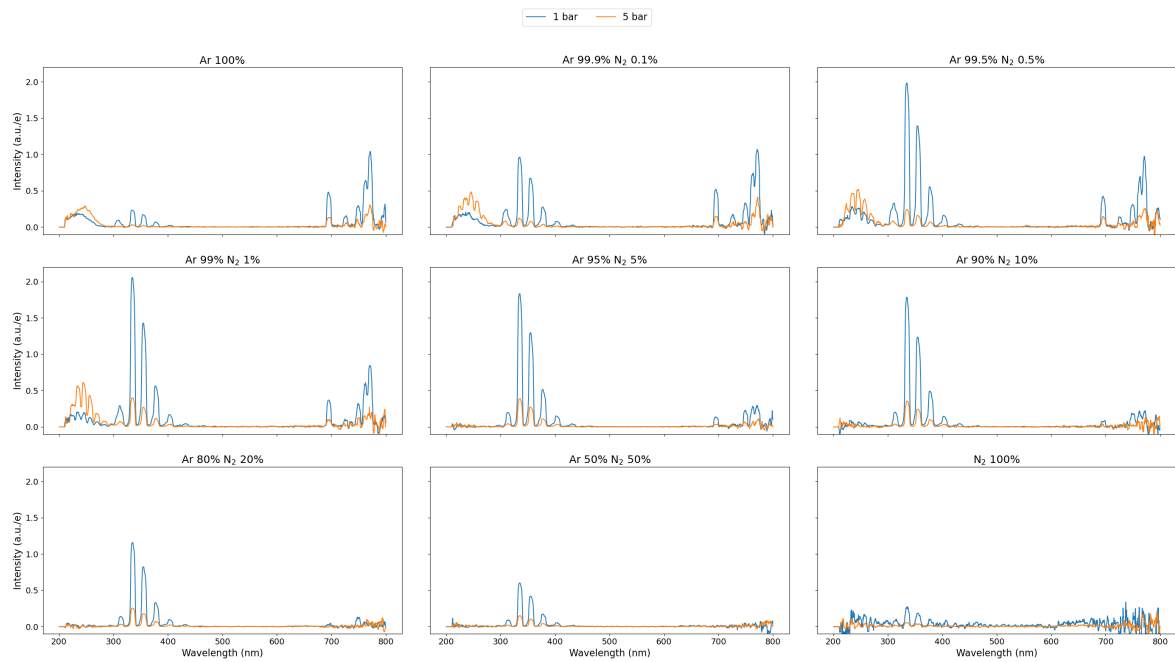
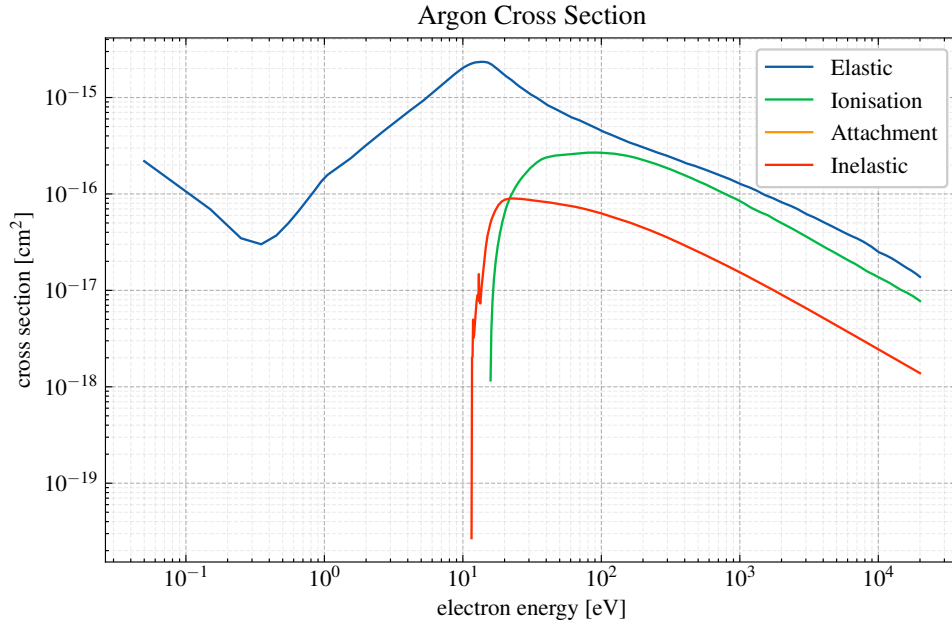
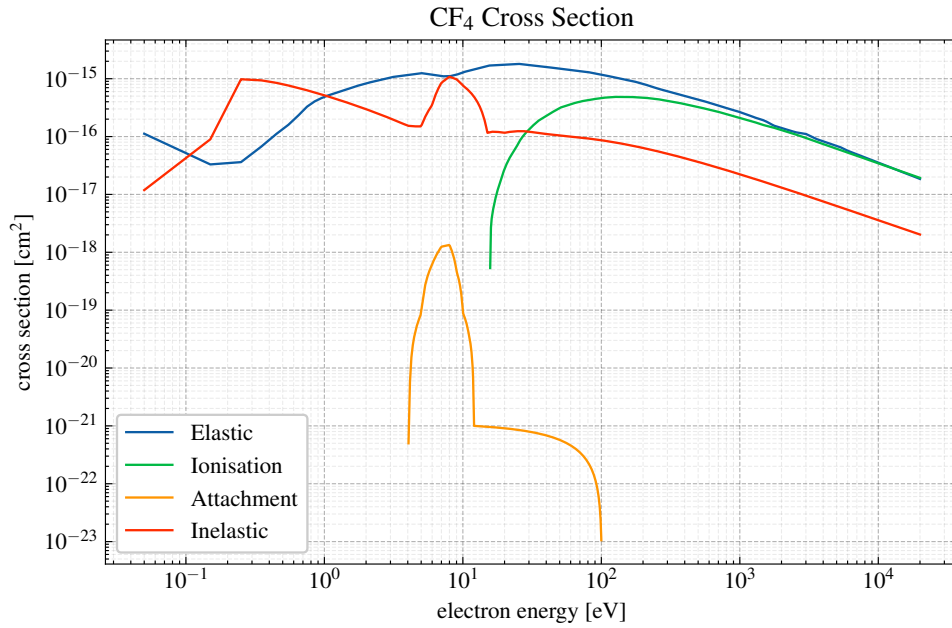


Figura 4: Caption

4. Modelo Cinético

4.1. Introducción a los modelos cinéticos

**Figura 5:** Ar cross sections**Figura 6:** CF₄ cross sections

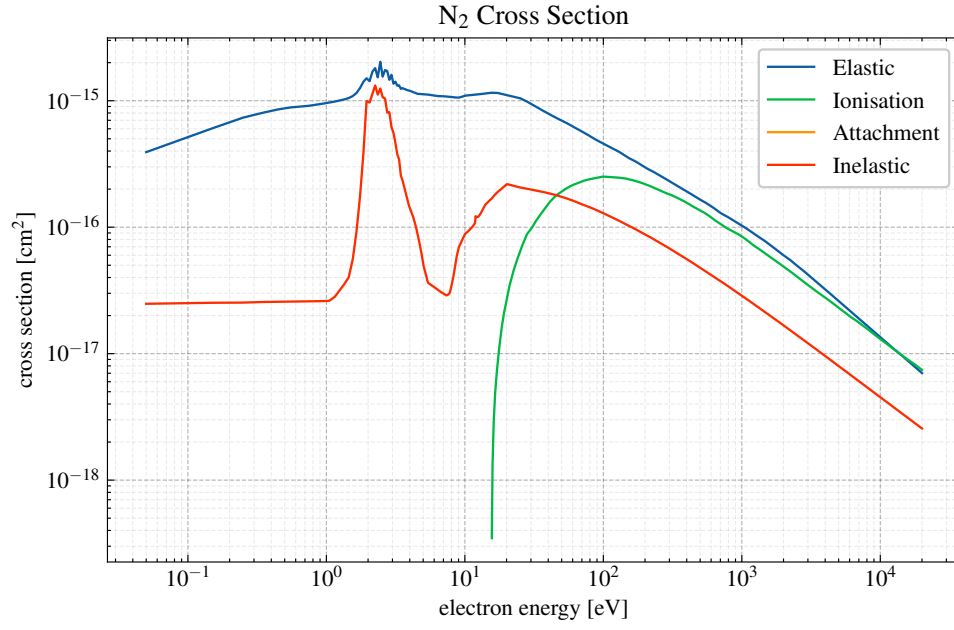
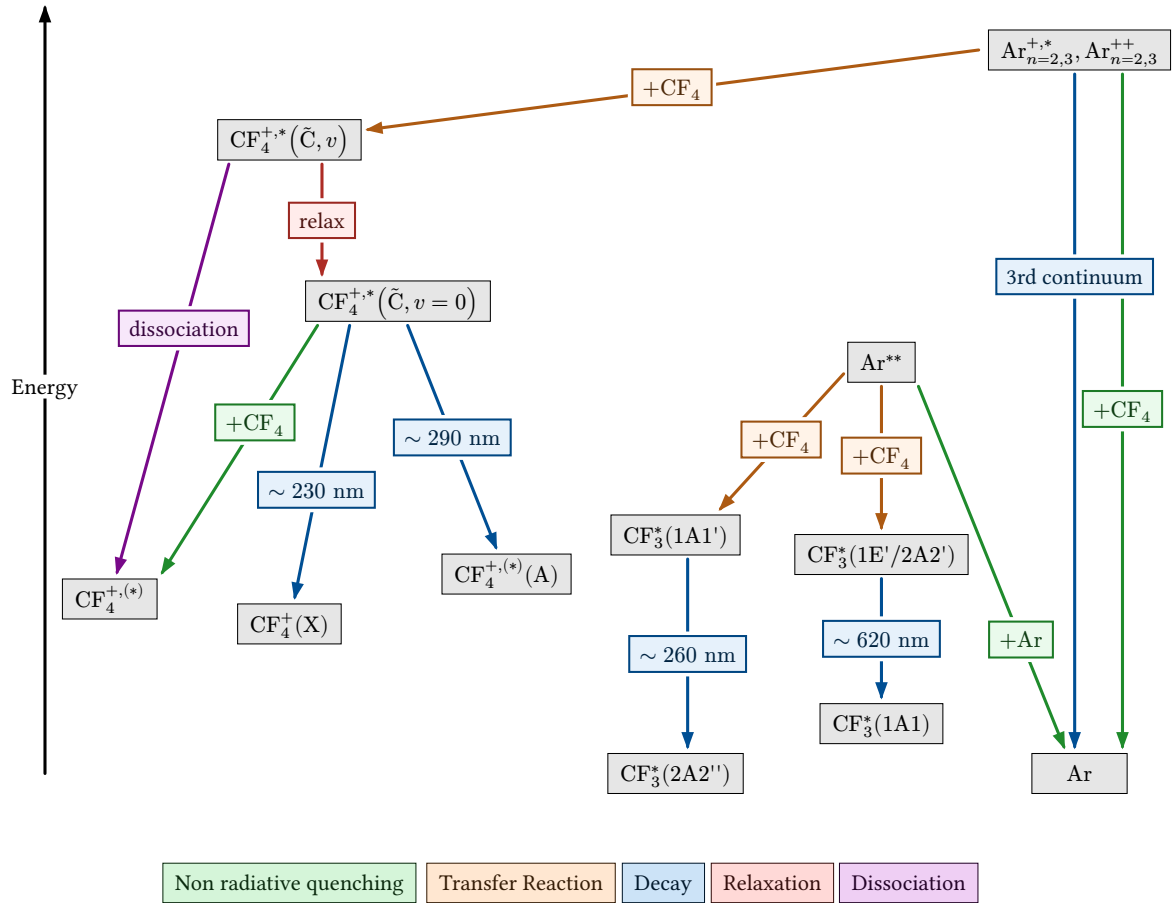
4.2. Ar – CF₄

Así pues, el centelleo de una mezcla de Ar – CF₄ tiene 3 componentes de emisión. La emisión del visible, por parte de la especie CF₃*:

$$Y_{\text{CF}_3, \text{vis}} = N_{\text{CF}_3} \cdot P_{\text{CF}_3^*} \Big|_{\text{dir}} + N_{\text{Ar}^{**}} \cdot P_{\text{Ar}^{**}} \cdot \frac{n \cdot f \cdot K_{\text{Ar}^{**}Q(\text{CF}_4)}}{n \cdot f \cdot K_{\text{Ar}^{**}Q(\text{CF}_4)} + n \cdot (1 - f) \cdot K_{\text{Ar}^{**}Q(\text{Ar})} + 1/\tau_{\text{Ar}^{**}}} \quad (1)$$

La emisión del ultravioleta por la especie CF₄^{+,*}

$$Y_{\text{CF}_4, \text{vis}} = \cdot P_{\text{CF}_4^{+,*}} \cdot \left[N_{\text{CF}_3} \cdot P_{\text{CF}_4^{+,*}} \Big|_{\text{dir}} + N_{\text{Ar}^{++}} \cdot P_{\text{Ar}^{++}} \left(\frac{n \cdot f \cdot K_{\text{Ar}^{++}Q(\text{CF}_4)}}{1/\tau_{3\text{rd}} + n \cdot f \cdot K_{\text{Ar}^{++}Q(\text{CF}_4)}} \right) \right] \quad (2)$$

**Figure 7:** N2 cross sections**Figure 8:** ArCF4 kinetic model

$$P_{\text{CF}_4^{+,*}} = \frac{n \cdot f \cdot K_{\text{relax}}}{n \cdot f \cdot K_{\text{relax}} + 1/\tau_{\text{dissc}}} \cdot \frac{1/\tau_{\text{uv}}}{1/\tau_{\text{uv}} + n \cdot f \cdot K_{\text{CF}_4^{+,*}Q(\text{CF}_4)}} \quad (3)$$

y la emisión del ultravioleta por la especie $\text{Ar}_{3\text{rd}}$:

$$Y_{\text{Ar}_{3\text{rd}},\text{uv}} = \cdot N_{\text{Ar}^{++}} \cdot \left(\frac{1/\tau_{3\text{rd}}}{1/\tau_{3\text{rd}} + n \cdot f \cdot K_{\text{Ar}^{++}Q(\text{CF}_4)}} \right) \quad (4)$$

4.3. Ar – N₂

Así pues, el centelleo de una mezcla de Ar – N₂ viene dada por

$$Y_{N_2} = P_{N_2^*(C)} \cdot \left(N_{N_2^*} \cdot P_{N_2} + N_{Ar^*(s_4)} \cdot P_{Ar^*(s_4)} + N_{Ar^*(s_5)} \cdot P_{Ar^*(s_5)} \right) \quad (5)$$

$$P_{N_2^*(C)} = \frac{1/\tau_{N_2^*(C)}}{1/\tau_{N_2^*(C)} + n \cdot f \cdot K_{N_2^*(C)Q(N_2)} + n \cdot (1-f) \cdot K_{N_2^*(C)Q(Ar)}} \quad (6)$$

$$P_{Ar^*(s_5)} = \frac{n \cdot f \cdot K_{Ar^*(s_5)Q(N_2^*(C))}}{n \cdot f \cdot K_{Ar^*(s_5)Q(N_2^*(C))} + n^2 \cdot f \cdot K_{Ar^*(s_5)Q(2Ar)}} \quad (7)$$

$$P_{Ar^*(s_4)} = \frac{n \cdot f \cdot K_{Ar^*(s_4)Q(N_2^*(C))}}{n \cdot f \cdot K_{Ar^*(s_4)Q(N_2^*(C))} + n^2 \cdot f \cdot K_{Ar^*(s_4)Q(2Ar)}} \quad (8)$$

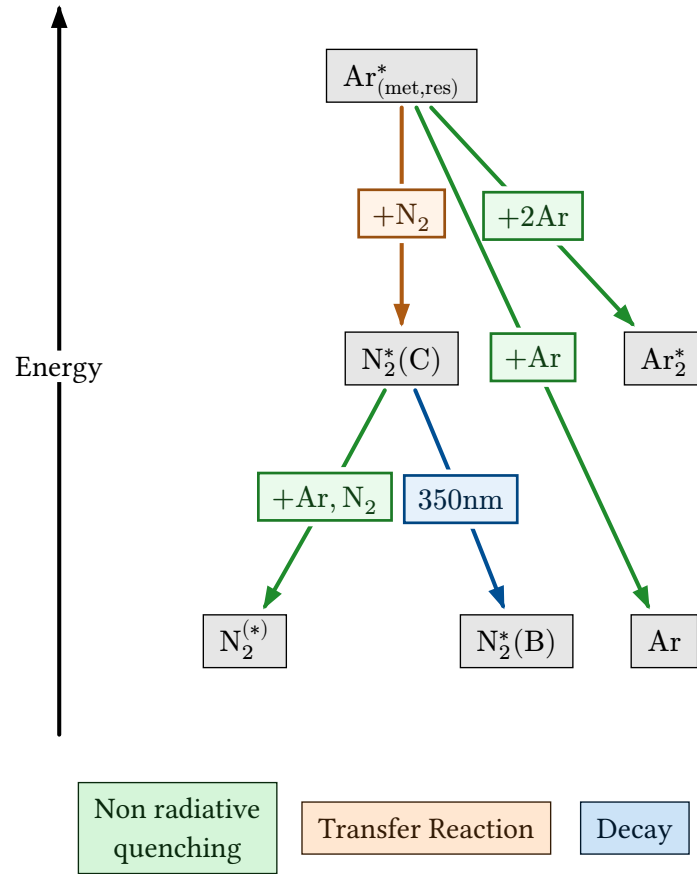


Figura 9: ArN₂ kinetic model

Parameter	[2]	[4]	[1]	[6]	[5]	Other
$1/\tau_{N_2^+(C)}$	$2.6 \times 10^7 \text{ s}^{-1}$	$2.07 \times 10^7 \text{ s}^{-1}$	$3.3 \times 10^7 \text{ s}^{-1}$ $2.5 \times 10^7 \text{ s}^{-1}$	$2.74 \times 10^7 \text{ s}^{-1}$	$2.66 \times 10^7 \text{ s}^{-1}$	
$k_{N_2^+(C)Q(N_2^*)}$	$7.1 \times 10^{-18} \text{ m}^3 \text{ s}^{-1}$	$1.12 \times 10^{-17} \text{ m}^3 \text{ s}^{-1}$	$10^{-17} \text{ m}^3 \text{ s}^{-1}$	$1.4 \times 10^{-17} \text{ m}^3 \text{ s}^{-1}$	$10^{-18} \text{ m}^3 \text{ s}^{-1}$	
$k_{N_2^+(C)Q(Ar)}$	-	$5.6 \times 10^{-19} \text{ m}^3 \text{ s}^{-1}$	-	$8.6 \times 10^{-19} \text{ m}^3 \text{ s}^{-1}$	$5.6 \times 10^{-19} \text{ m}^3 \text{ s}^{-1}$	
$k_{Ar^+(1s5)Q(N_2^+(C))}$	$3.2 \times 10^{-17} \text{ m}^3 \text{ s}^{-1}$	$3.0 \times 10^{-17} \text{ m}^3 \text{ s}^{-1}$	-	-	$1.1 \times 10^{-17} \text{ m}^3 \text{ s}^{-1}$	
$k_{Ar^+(1s5)Q(N_2)}$	-	-	-	-	$1.26 \times 10^{-17} \text{ m}^3 \text{ s}^{-1}$	
$k_{Ar^+(1s5)Q(2Ar)}$	-	-	-	-	$10^{-44} \text{ m}^6 \text{ s}^{-1}$	$7.93 \times 10^6 \text{ s}^{-1}$
$k_{Ar^+(1s4)Q(N_2^+(C))}$	-	$3.0 \times 10^{-17} \text{ m}^3 \text{ s}^{-1}$	$1.5 \times 10^{-17} \text{ m}^3 \text{ s}^{-1}$ $3.6 \times 10^{-17} \text{ m}^3 \text{ s}^{-1}$	-	-	
$k_{Ar^+(1s4)Q(N_2)}$	-	-	$3 \times 10^{-17} \text{ m}^3 \text{ s}^{-1}$ $3.6 \times 10^{-17} \text{ m}^3 \text{ s}^{-1}$	-	-	
$k_{Ar^+(1s4)Q(2Ar)}$	-	-	$10^{-44} \text{ m}^6 \text{ s}^{-1}$	-	$10^{-44} \text{ m}^6 \text{ s}^{-1}$	$9.24 \times 10^5 \text{ s}^{-1}$

Tabla 1: Parameters in the literature

5. Simulaciones

5.1. Degrad

5.2. Garfield++

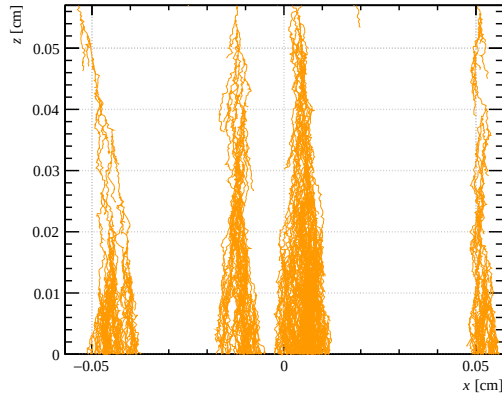


Figura 10: Caption

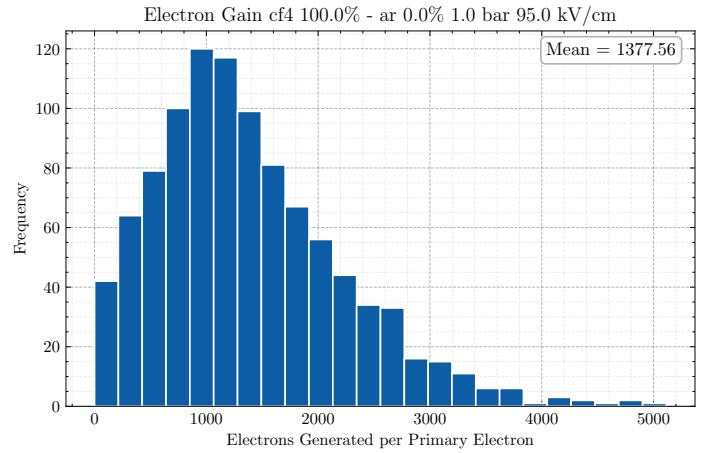


Figura 11: Caption

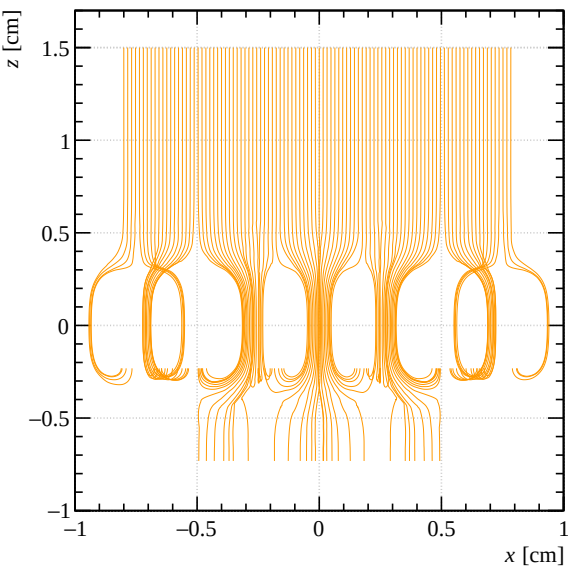


Figura 12: Caption

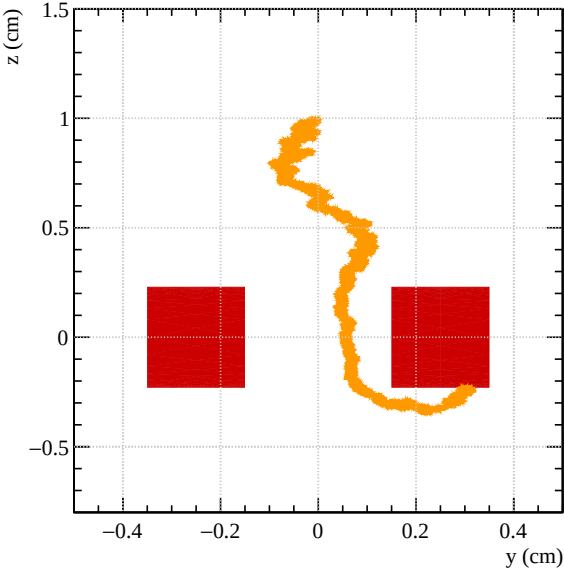


Figura 13: Caption

6. Obtención de Parámetros

6.1. Ar – CF₄

Las ecuaciones del ajuste son:

$$Y_{\text{CF}_3, \text{vis}} = \frac{N_{\text{norm}}}{W_I(f)} \left(N_{\text{CF}_3} \cdot P_{\text{CF}_3} \Big|_{\text{dir}} + N_{\text{Ar}^{**}} \cdot P_{\text{Ar}^{**}} \cdot \frac{n \cdot f \cdot K_{\text{Ar}^{**}Q(\text{CF}_4)}}{n \cdot f \cdot K_{\text{Ar}^{**}Q(\text{CF}_4)} + n \cdot (1 - f) \cdot K_{\text{Ar}^{**}Q(\text{Ar})} + 1/\tau_{\text{Ar}^{**}}} \right)$$

$$Y_{\text{CF}_4, \text{vis}} = \frac{N_{\text{norm}}}{W_I(f)} \cdot P_{\text{CF}_4^{+,*}} \cdot \left[N_{\text{CF}_3} \cdot P_{\text{CF}_4^{+,*}} \Big|_{\text{dir}} + N_{\text{Ar}^{++}} \cdot P_{\text{Ar}^{++}} \cdot \left(\frac{n \cdot f \cdot K_{\text{Ar}^{++}Q(\text{CF}_4)}}{1/\tau_{3\text{rd}} + n \cdot f \cdot K_{\text{Ar}^{++}Q(\text{CF}_4)}} \right) \right]$$

$$P_{\text{CF}_4^{+,*}} = \frac{n \cdot f \cdot K_{\text{relax}}}{n \cdot f \cdot K_{\text{relax}} + 1/\tau_{\text{disoc}}} \cdot \frac{1/\tau_{\text{uv}}}{1/\tau_{\text{uv}} + n \cdot f \cdot K_{\text{CF}_4^{+,*}Q(\text{CF}_4)}}$$

$$Y_{\text{Ar}_{3\text{rd}}, \text{vis}} = \frac{N_{\text{norm}}}{W_I(f)} \cdot N_{\text{Ar}^{++}} \cdot \left(\frac{1/\tau_{3\text{rd}}}{1/\tau_{3\text{rd}} + n \cdot f \cdot K_{\text{Ar}^{++}Q(\text{CF}_4)}} \right)$$

Tabla 2: Parámetros obtenidos del ajuparamste global.

Parámetro	Valor	Incertidumbre	Incertidumbre (%)
N_{norm}	0.2044	0.1958	95.79
$P_{\text{CF}_3} \Big _{\text{vis, dir}}$	0.1019	0.097 81	95.98
$P_{\text{Ar}^{**}}$	0.2085	0.1999	95.87
$K_{\text{Ar}^{**}, Q(\text{Ar})} [\text{ns}]$	0.2475	0.1944	78.54
$1/\tau_{\text{disoc}} K_{\text{relax}}$	3.046×10^{-3}	2.588×10^{-3}	84.96
$\tau_{\text{uv}} K_{\text{CF}_4^{+,*}Q(\text{CF}_4)}$	0.065	0.056 56	87.02
$P_{\text{CF}_4^{+,*}} \Big _{\text{dir}}$	0.3347	0.3143	93.88
$K_{\text{Ar}^{++}, Q(\text{CF}_4)} [\text{ns}]$	50	234.7	469.4
$P_{\text{Ar}^{++}}$	0.4813	0.4935	102.5
$K_{\text{Ar}^{**}, Q(\text{CF}_4)} [\text{ns}]$	4.857	3.633	74.8
$P_{\text{CF}_3} \Big _{\text{uv, dir}}$	10^{-3}	0.29	2.9×10^4

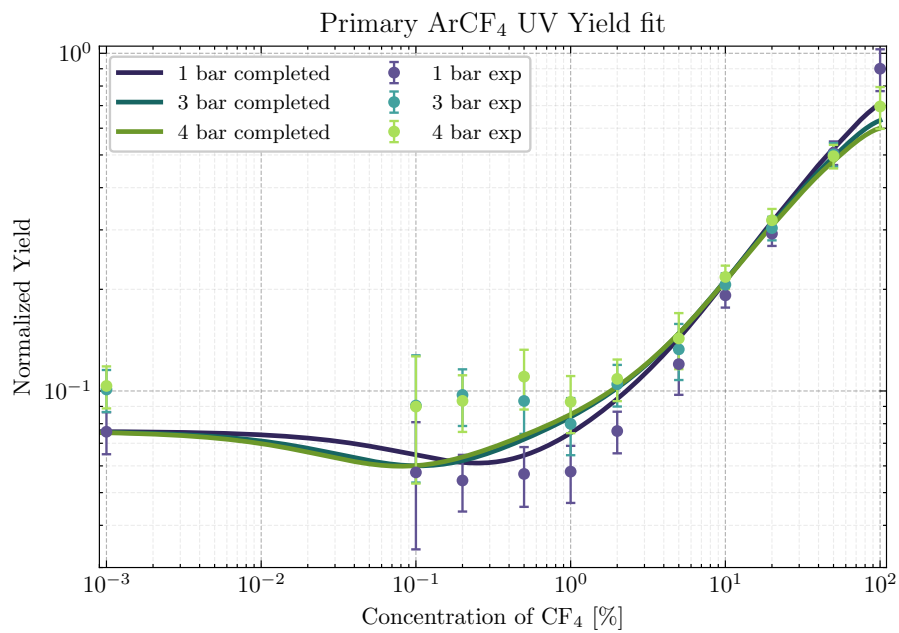
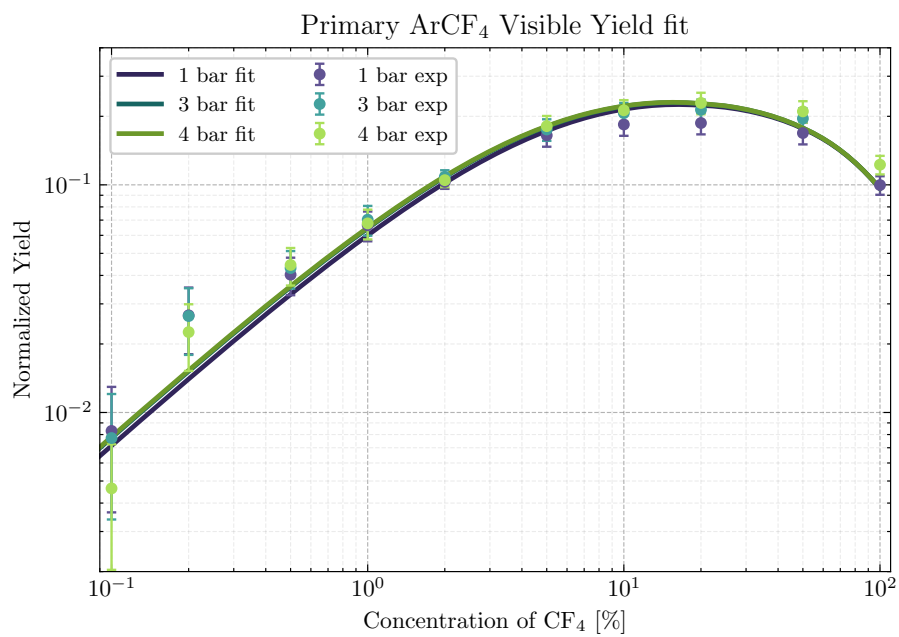
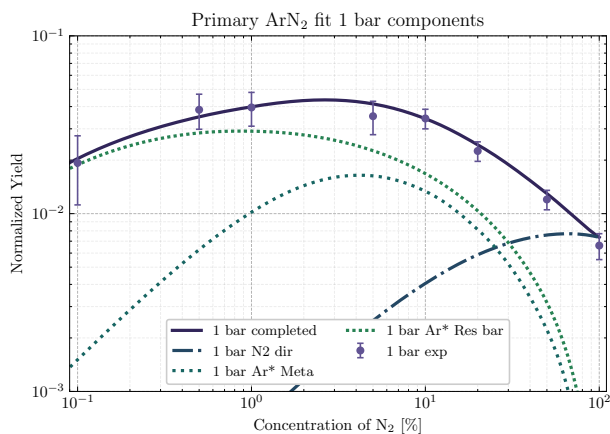
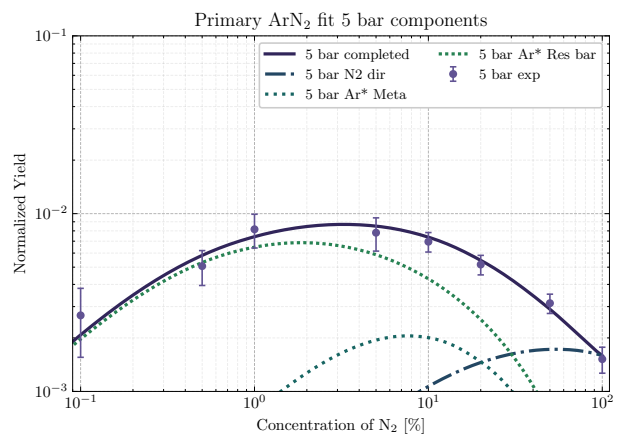
6.2. Ar – N₂

$$Y_{\text{N}_2} = \frac{N_{\text{norm}}}{W_I(f)} \cdot P_{\text{N}_2^*(\text{C})} \cdot \left(N_{\text{N}_2^*} \cdot P_{\text{N}_2} + N_{\text{Ar}^*(s_4)} \cdot P_{\text{Ar}^*(s_4)} + N_{\text{Ar}^*(s_5)} \cdot P_{\text{Ar}^*(s_5)} \right) \quad (9)$$

$$P_{\text{N}_2^*(\text{C})} = \frac{1/\tau_{\text{N}_2^*(\text{C})}}{1/\tau_{\text{N}_2^*(\text{C})} + n \cdot f \cdot K_{\text{N}_2^*(\text{C})Q(\text{N}_2)} + n \cdot (1 - f) \cdot K_{\text{N}_2^*(\text{C})Q(\text{Ar})}} \quad (10)$$

$$P_{\text{Ar}^*(s_5)} = \frac{n \cdot f \cdot K_{\text{Ar}^*(s_5)Q(\text{N}_2^*(\text{C}))}}{n \cdot f \cdot K_{\text{Ar}^*(s_5)Q(\text{N}_2^*(\text{C}))} + n^2 \cdot f \cdot K_{\text{Ar}^*(s_5)Q(2\text{Ar})}} \quad (11)$$

$$P_{\text{Ar}^*(s_4)} = \frac{n \cdot f \cdot K_{\text{Ar}^*(s_4)Q(\text{N}_2^*(\text{C}))}}{n \cdot f \cdot K_{\text{Ar}^*(s_4)Q(\text{N}_2^*(\text{C}))} + n^2 \cdot f \cdot K_{\text{Ar}^*(s_4)Q(2\text{Ar})}} \quad (12)$$

**Figura 14:** Caption**Figura 15:** Caption**Figura 19:** Caption**Figura 20:** Caption

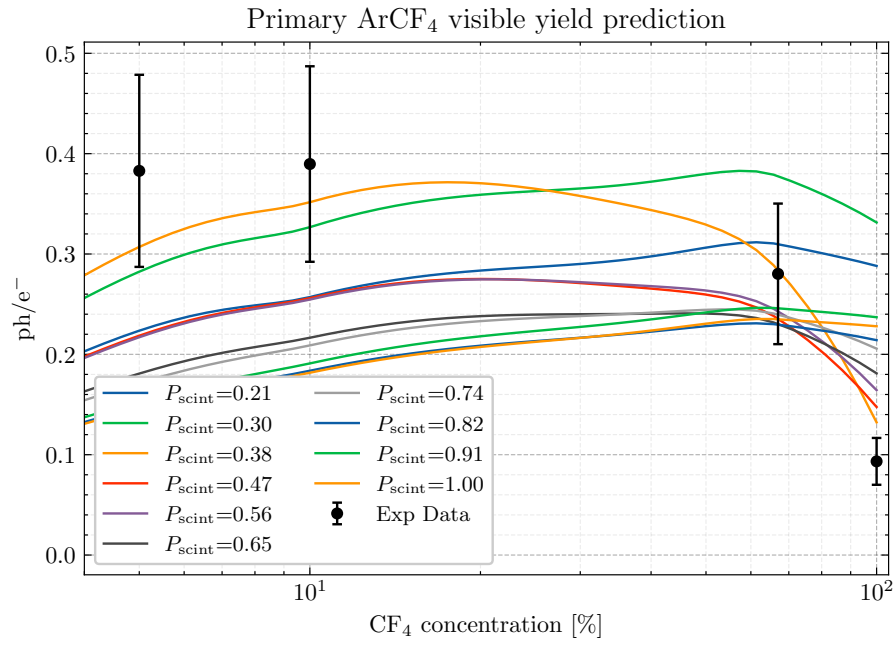
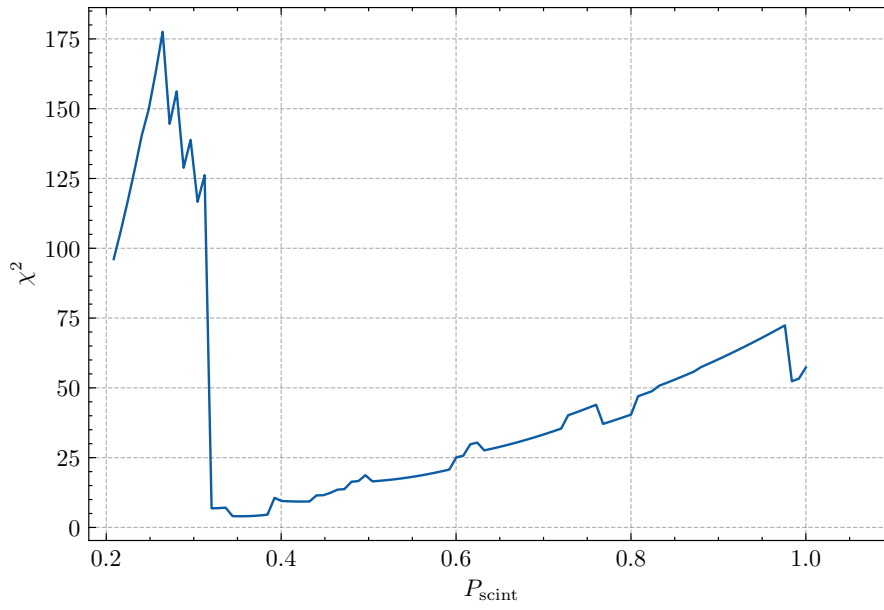
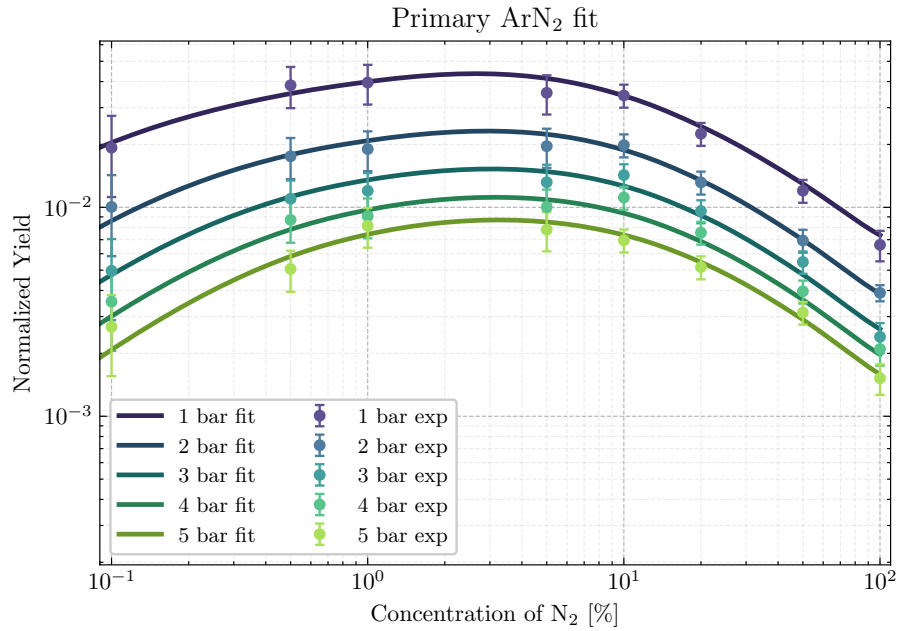
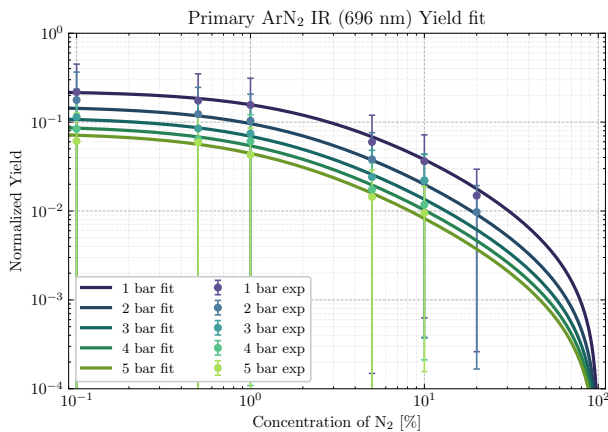
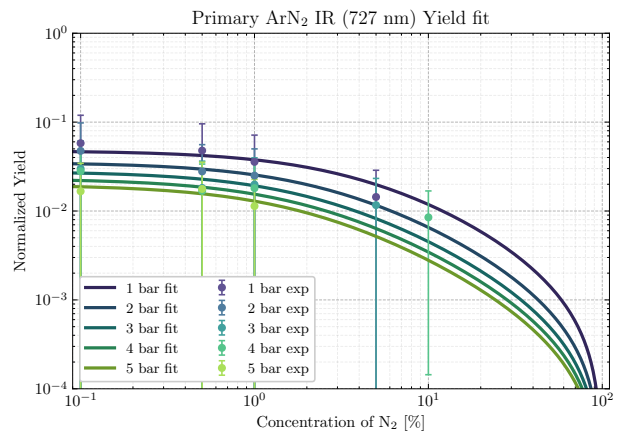
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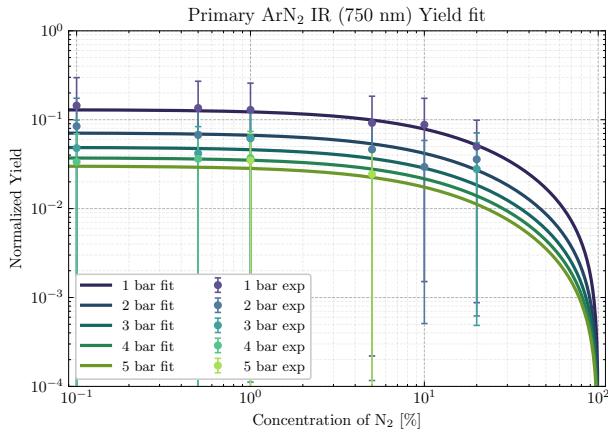
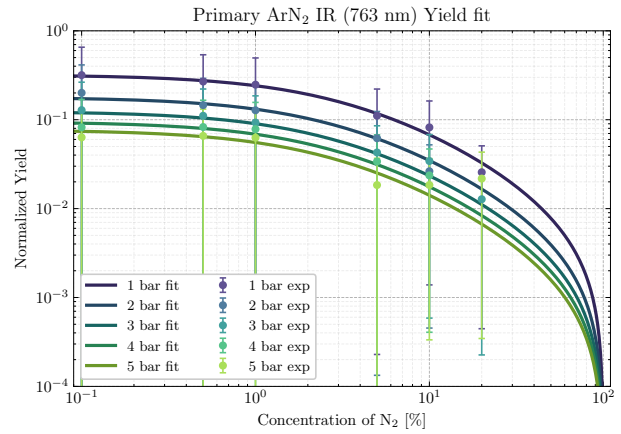
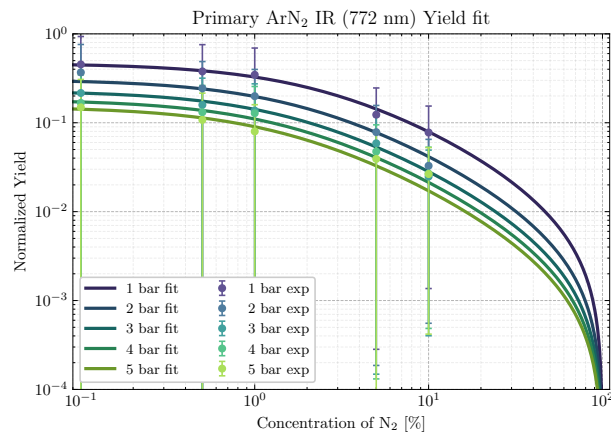
Tabla 3: Parameters known (x_0) and the obtained in the fit with a parameter factor freedom

Parameter	x_0	Factor x1.2	Factor x1.5	Factor x2	Factor x3
N_{norm}	0.250	0.150	0.150	0.150	0.172
P_{N_2}	0.00	0.177	0.115	0.0684	0.0564
τ_{N_2} [ns]	37.8	45.4	52.0	40.8	43.4
$K_{N_2Q(N_2)}$ [ns ⁻¹]	0.284	0.341	0.188	0.142	0.126
$K_{N_2Q(Ar)}$ [ns ⁻¹]	0.0191	0.0229	0.0286	0.0382	0.0366
$K_{Ar_{1s5}Q(N_2(C))}$ [ns ⁻¹]	0.655	0.524	0.432	0.327	0.342
$K_{Ar_{1s5}Q(N_2(B))}$ [ns ⁻¹]	0.0430	0.0516	0.0284	0.0861	0.127
$K_{Ar_{1s5}Q(2Ar)}$ [ns ⁻¹]	7.93×10^{-3}	9.52×10^{-3}	0.0119	6.70×10^{-3}	6.07×10^{-3}
$K_{Ar_{1s4}Q(N_2(C))}$ [ns ⁻¹]	0.686	0.549	0.453	0.431	0.364
$K_{Ar_{1s4}Q(N_2(B))}$ [ns ⁻¹]	0.202	0.242	0.303	0.403	0.515
$K_{Ar_{1s4}Q(2Ar)}$ [ns ⁻¹]	9.24×10^{-4}	1.11×10^{-3}	6.27×10^{-4}	4.62×10^{-4}	4.02×10^{-4}

**Figura 18:** Caption

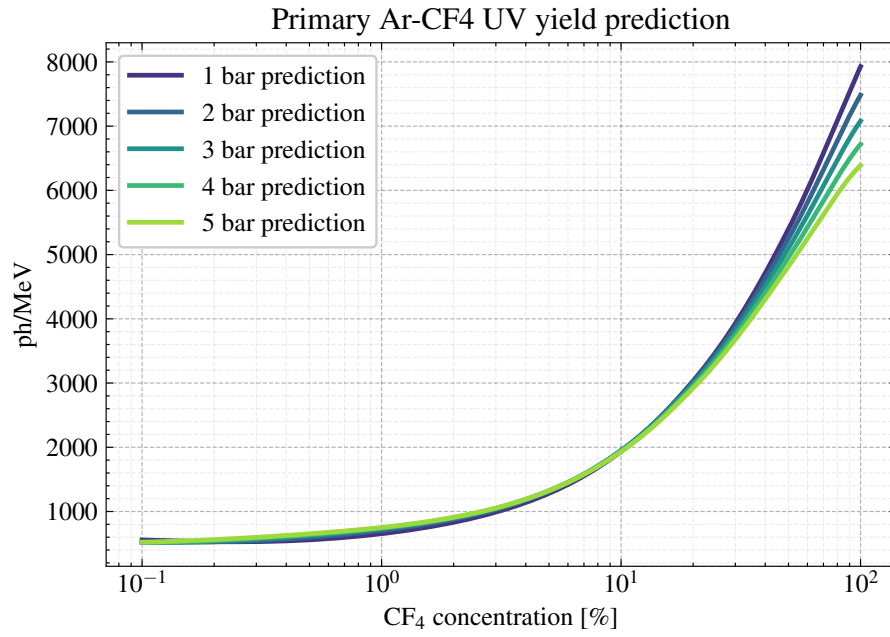
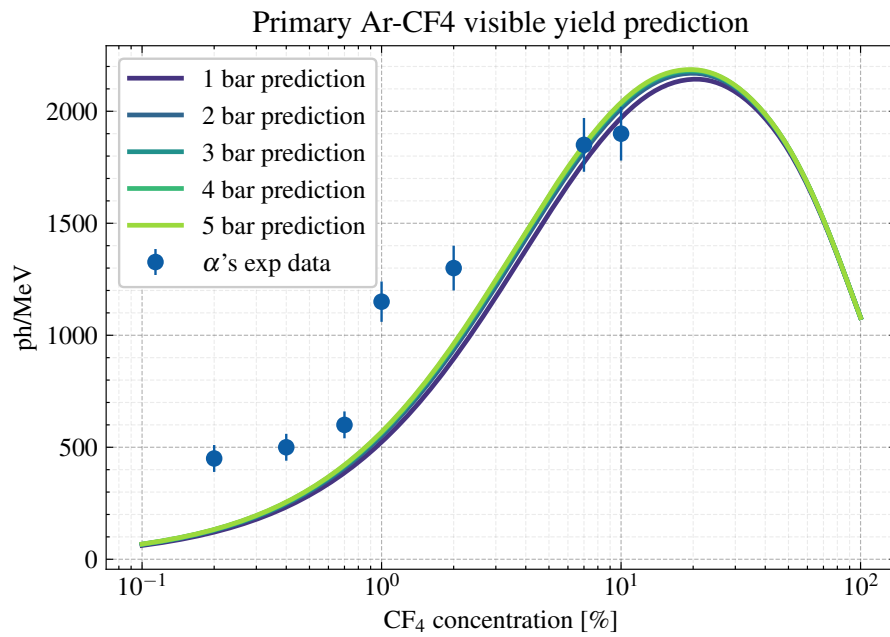
6.3. Ar Infrarrojo

**Figura 21:** Caption**Figura 22:** Caption

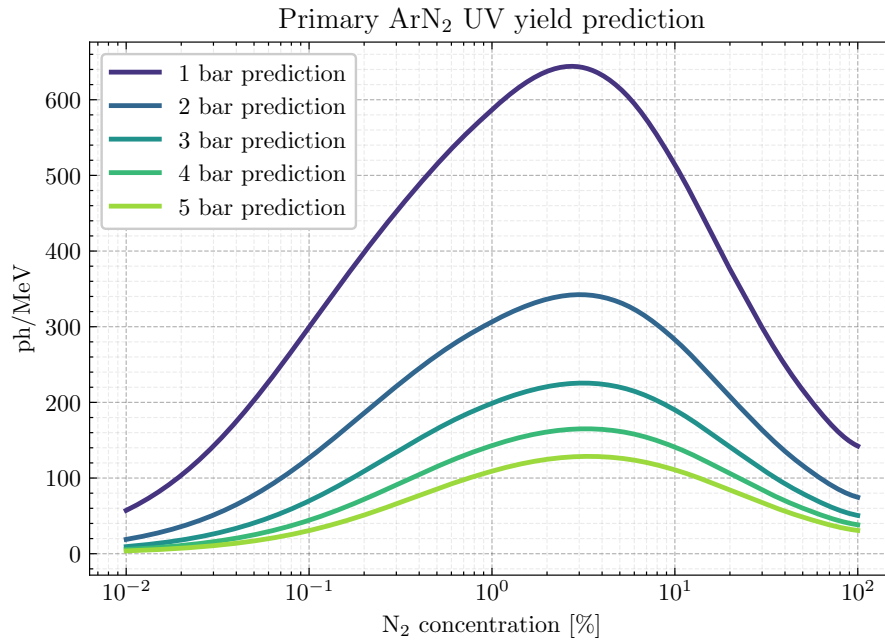
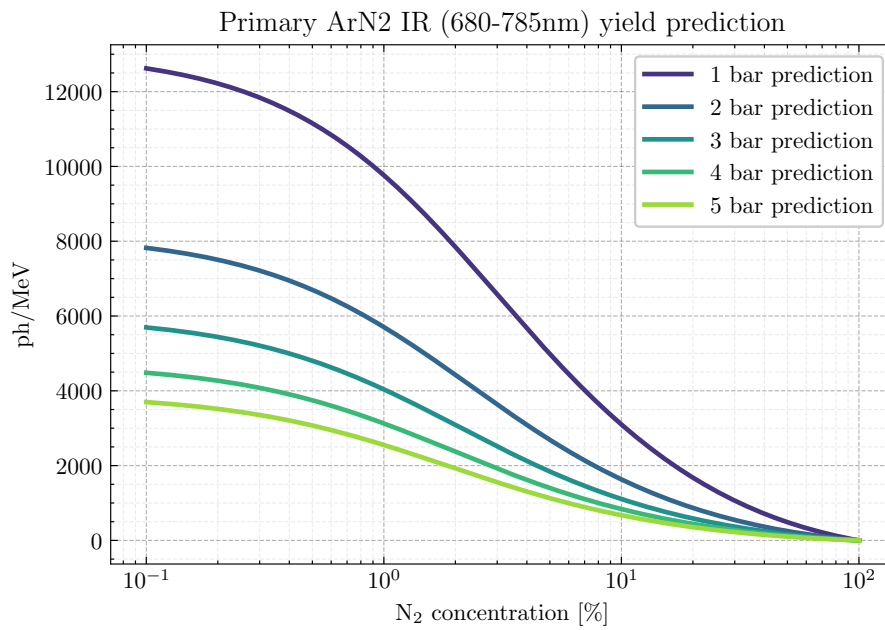
**Figura 23:** Caption**Figura 24:** Caption**Figura 25:** Caption

7. Predicciones

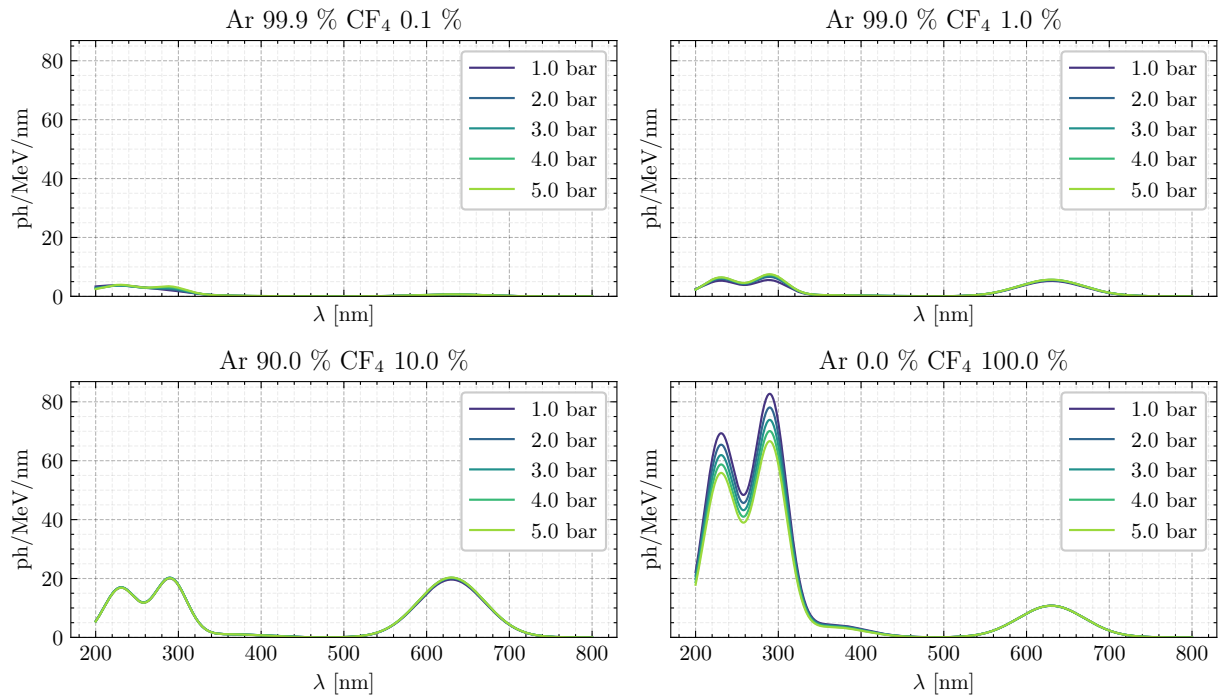
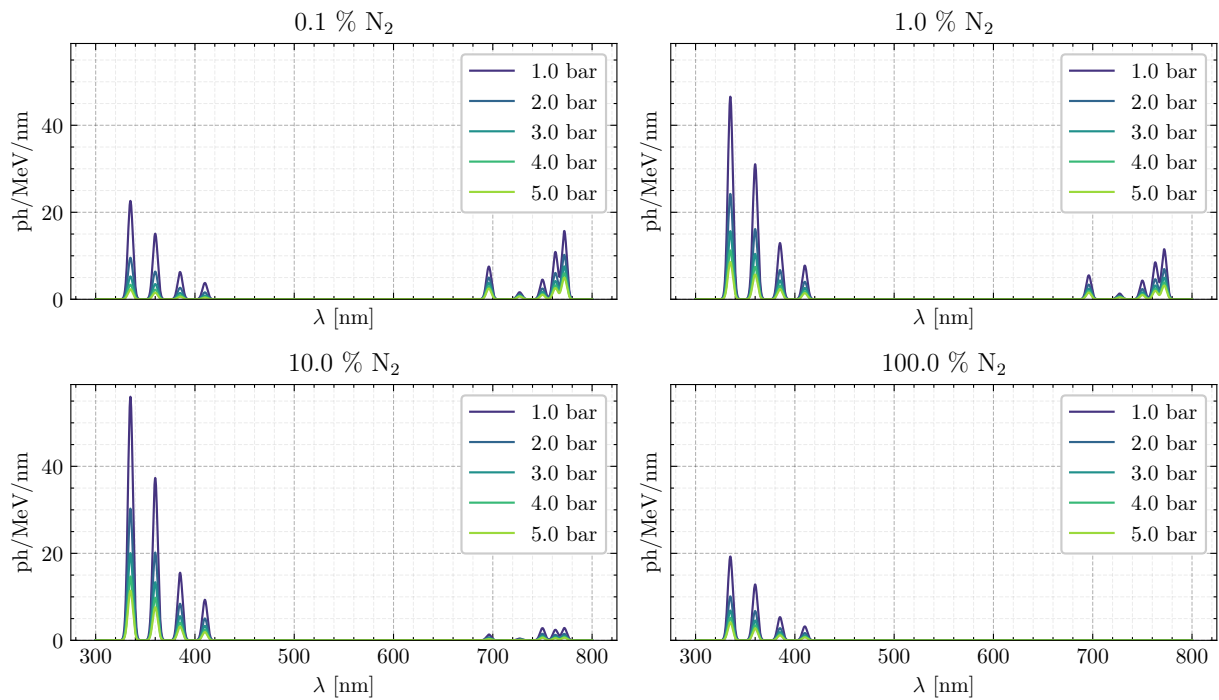
7.1. Predicciones en el primario

**Figura 26:** Caption**Figura 27:** Caption

7.2. Predicciones en el secundario

**Figura 28:** Caption**Figura 29:** Caption

7.3. Espectros

Primary Ar-CF₄ Spectra Prediction**Figura 30:** CaptionPrimary Ar-N₂ Spectra Prediction**Figura 31:** Caption

8. Aplicaciones

8.1. *Photon-Feedback*

8.2. Simulaciones en alta ganancia

Referencias

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